

QuickSort Algorithm using Lomuto Partition Scheme

QuickSort Algorithm (Lomuto Partition)

Algorithm 1 QuickSort(A , $left$, $right$)

```
1: if  $left < right$  then  
2:    $p \leftarrow \text{PARTITION}(A, left, right)$   
3:   QUICKSORT( $A, left, p - 1$ )  
4:   QUICKSORT( $A, p + 1, right$ )  
5: end if
```

Partition Procedure (Lomuto)

Algorithm 2 Partition(A , $left$, $right$)

```
1:  $pivot \leftarrow A[right]$   
2:  $i \leftarrow left - 1$   
3: for  $j \leftarrow left$  to  $right - 1$  do  
4:   if  $A[j] \leq pivot$  then  
5:      $i \leftarrow i + 1$   
6:     swap  $A[i] \leftrightarrow A[j]$   
7:   end if  
8: end for  
9: swap  $A[i + 1] \leftrightarrow A[right]$   
10: return  $i + 1$ 
```
